

PRESENT AND FUTURE TYPES OF FAST BREEDER REACTORS

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1. Introduction

The massive improvement in uranium utilization attainable by the fast breeder reactor compared with any thermal reactor system will be of increasing importance in the future as it appears that the world's reserves of high-grade ores are comparatively limited. It is this feature of the fast breeder reactor that assures it a very special place in the long-term power programme of any country that currently depends on coal, oil or gas fuels which has assessed its energy requirements far enough ahead. The position that the fast breeder system occupies in short-term national programmes is, on the other hand, more variable since it depends on the present energy resources available to a particular country and, in some cases, on a judgement of the time needed to develop the required technology.

In the early 1950's, the United Kingdom considered that it could be faced with an early shortage of conventional fuels. As a result, it initiated what is still the world's largest programme to develop nuclear power when judged on the output of the reactor plant. This programme was based on thermal reactors to be complemented subsequently by the introduction of fast breeder reactors as they became available to utilize the plutonium produced in these thermal reactors. In order to achieve this broad programme, the construction of an experimental fast breeder reactor using sodium cooling (the DFR) was begun at Dounreay almost at the same time as that of the first thermal systems. Although it became apparent that forecasts of a serious shortfall in conventional fuels were premature, the British nuclear power programme continued with the building of a considerable number of large thermal reactor power stations. DFR has been in successful operation since it was commissioned in 1959 (Fig. 1) and, more recently, sanction has been given for a 250 MW(e) prototype fast reactor also to be built at Dounreay. By 1975, therefore, the United Kingdom should be in a position to embark upon the second phase of its nuclear power programme, a combination of

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thermal and fast reactors, which calls for the construction of large output, fast reactor power stations at an increasing rate. While this second phase would not be possible without the initial stage, its consequences upon the economy of the nation will be much more profound because of the anticipated scale of installation.

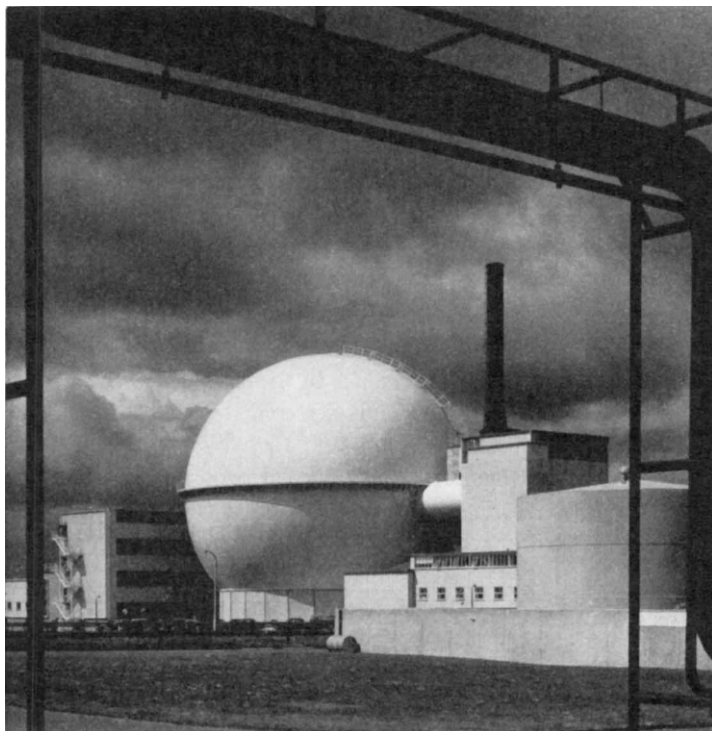


FIG. 1. View of Dounreay fast reactor.

However, it necessarily remains to be unequivocally demonstrated that the components of the sodium system can be manufactured within the estimated prices, that an economic fuel burnup can be achieved and that the reactor can be operated in a safe and reliable manner. Nor can it be affirmed at present—and it would be unwise to assume it—that the sodium cooled fast reactor will occupy prime place indefinitely unchallenged by other possible fast systems. It is for this reason that the UKAEA has engaged in investigations of alternative types.

2. The Sodium Cooled Fast Reactor

There is now little doubt that the sodium cooled fast reactor is technically feasible for large power units and current estimates indicate that a programme

of such reactors would show substantial savings compared with complete reliance on conventional and thermal reactor stations. The announcement by the Minister of Technology on February 9, 1966, that the United Kingdom is to proceed with the construction of a 250 MW(e) prototype fast reactor confirms that this type of reactor represents the next stage in its developing nuclear power programme.

Turning from the specific circumstances of the British programme, and considering current world opinions on sodium cooled fast reactors, there seems to be remarkable uniformity of view on the parameters and general design for a reactor of this type. Working to a large extent independently, designers in the U.S.S.R., in the U.S.A. and in a number of European countries are putting forward, at least for immediate development, choices of both fuel and system configuration that have much in common.

Commercial fast reactors have been forecast in several countries as going into service at various dates from 1975 onwards. The outputs to be obtained from these reactors will depend upon the capacity and stability characteristics of the electrical network to which they will be connected. However, there seems to be a general consensus of opinion that 1000 MW(e) constitutes an appropriate size for early fast reactors provided these are associated with substantial electrical distribution systems. Subsequent developments are difficult to predict but there would appear to be no reason why units of 2000 MW(e) or even more should not be practicable within the first decade following their introduction. With increasing output comes the prospect of significant savings in capital cost and some promise of improvement in neutron economy. Improvements in neutron economy might, however, be diminished or nullified if special design features have to be introduced to counter the tendency for void coefficients of reactivity to become more positive as the size of the reactor core increases.

A wide range of fuels is possible but, at the present time, it is only the oxide form on which there is substantial supporting evidence for burning plutonium. The trend is, however, towards improving breeding ratios which could favour the development of carbide fuels to obtain the beneficial effect of increased fuel density and reduced moderation. Alternative possibilities are oxy-carbides and nitrides which have potential manufacturing or performance advantages, although substantially less is currently known of their characteristics. There may also be some resurgence in interest in metallic fuels. The effect of improved breeding ratio needs to be considered together with advantages to be obtained from longer burnup and higher fuel ratings and these aspects are certain to have a bearing on the ultimate choice of fast reactor fuels.

To improve breeding ratio, it is necessary to reduce coolant and cladding core fractions, while to reduce fuel fabrication costs, the trend is to larger fuel assemblies. These aspects lead to consideration of fuel shapes other than

circular rods, and arrangements such as annular or shell-and-tube geometries are receiving attention. The concept of venting released fission product gases is particularly relevant to complex fuel geometries. At the present time, almost all work on cladding that is in an advanced stage is focused on stainless steel. However, nickel based materials offer the prospects of enhanced strength at high temperature in the longer term and the high temperature properties of molybdenum and its alloys are also attractive if fabrication problems and ductility limitations can be overcome, and the adverse nuclear effects accommodated.

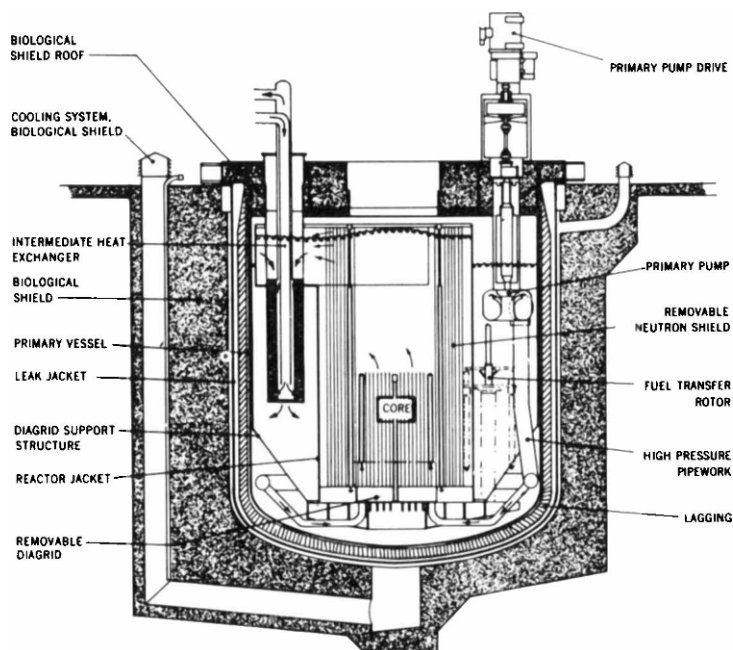


FIG. 2. Sodium cooled reactor—single tank arrangement.

It is always difficult to forecast trends of engineering design and this difficulty certainly applies to sodium cooled fast reactor design. At the present time, both single-tank schemes with primary coolant circulation pumps and intermediate heat exchangers in the primary tank (Fig. 2) and multi-tank schemes where these components are separately housed have been put forward as a means of obviating the loss-of-coolant accident. At the moment, there is no clear evidence that, in the longer term, one scheme will invariably prove to be substantially more economic than the other. Instead, it seems probable that both types will be successfully developed. Inevitably, many refinements and improvements in design can be expected and it is also

anticipated that further simplification will take place in containment concepts, in associated protective plant installed for reasons of safety, and in fuel handling equipment.

While the adoption of a single liquid metal coolant circuit (i.e. where the heat is transferred to steam from the primary coolant) would appear to offer the prospect of a simpler and more economical arrangement than the present designs incorporating an intermediate liquid metal, heat transport circuit, it has to be noted that not even the boldest designers have, as yet, pressed the claims of such single circuit designs. Several approaches to alternatives to the present double circuit designs can nevertheless be identified. Of these, the single wall steam generator in a single circuit may seem to be simplest in concept but it clearly presents most problems to the engineer in his task of providing adequate assurance of safety. Some designers may prefer to adopt a double walled arrangement which, in its essence, is the reduction of the secondary circuit to a static, conducting fluid between two walls in the steam generator. It is also possible to contemplate the use of secondary heat transfer media other than liquid metal and hydrocarbons such as benzene. However, most alternative secondary fluids have some limitation which could preclude them from showing any advantage over current arrangements.

Turning to the materials for the reactor itself, stainless steel is almost invariably chosen in present designs for the primary coolant circuit. In the secondary circuit, the possibility of stress corrosion in the steam generators has led some designers to adopt low alloy, ferritic steels. However, low alloy steels such as 2% chromium, 1% molybdenum have a tendency to decarburize in the presence of sodium. This is particularly damaging in mixed ferritic-stainless steel systems since mass transfer of the carbon takes place to the stainless steel which thereby becomes carburized. However, stabilized forms of 2% chromium stainless steel have already been developed which have substantially higher resistance to decarburization in sodium and it would be reasonable to assume that there will be further development of a comparable nature with other ferritic steels. Much of this development will have to be concentrated on improvements in their properties at high temperatures in order to accommodate the higher sodium temperatures that the designers would wish to adopt in more advanced designs of reactor.

3. Other Forms of Liquid Metal Cooled Fast Reactors

Consideration has been given to the employment of metals other than sodium to cool the cores of fast breeder reactors, particularly in the context of eliminating the need for a secondary heat transport circuit. Typical of this work have been studies based on the use of lead since this is not chemically reactive with water, its properties which affect the neutron physics of fast reactors are good and, as a coolant, it would provide a very good margin to

boiling and so limit the formation of local voidage. However, there are uncertainties regarding its compatibility with the more commonly used materials of construction and it is generally accepted that austenitic steels containing nickel would be unsuitable. Ferritic alloys might be used although some limitations might be experienced in high temperature properties for fuel cladding applications.

From the engineering viewpoint, lead would be regarded as an inconvenient coolant since it has a high density and a high melting point. The high density creates structural problems and pumping power with lead coolant would inherently be greater than with sodium. A design to minimize this could, however, probably be devised without too much sacrifice of neutron economy. The high melting point of lead gives rise to problems in circuit operation and in heat exchanger design. The use of an alloy eutectic such as lead/magnesium or lead/bismuth would result in a lower melting point but other problems then arise. Once the use of lead as a coolant is contemplated, the possibility of a number of fast reactor variants with fluid fuel is raised and it is these variants that have attracted more attention than solid fuelled schemes with configurations similar to those associated with sodium cooling.

4. Gas Cooled Fast Reactor

The gas cooled fast reactor has been a recurring topic of interest, because it has been clear that, in principle, it offers possible advantages in neutron economy and hence breeding ratio. Figures as high as 1.6 have been quoted for the latter but it has to be demonstrated that good fuel ratings are attainable if there is not to be a fuel inventory penalty in comparison with sodium cooled reactors. To achieve the requisite fuel rating, heat transfer rates between the fuel can and the coolant must be high, relative to thermal gas cooled reactors, and this leads to the proposed use of correspondingly high coolant pressures to minimize pumping power. The resulting design and safety problems would, with a steel pressure vessel, be truly formidable. However, the satisfactory development of pre-stressed concrete pressure vessels has made a solution to these problems less remote, and, in consequence, there has recently been a revival of interest in this form of fast reactor.

The highly developed state of gas coolant technology in the U.K. has naturally led to the question of whether this could readily be extended to a fast reactor. At least three coolants can be considered: helium, sulphur dioxide, and carbon dioxide, but no sufficiently strong reason has emerged for departing from carbon dioxide in the U.K. With this coolant, a heat rating of about 100 W/g ($\int K d\theta = 40$) might be achieved with a coolant pressure of 1000 lb/in² and a nominal maximum can temperature of 700°C, with a pumping power of only 2.5% of the heat output. The fissile rating would

be about 600 W/g. The engineering of the primary circuit for such a plant could be closely based on current gas cooled reactor practice. The pressure vessel, blowers, and boilers would not involve any new technology, and the proper selection of constructional materials would be well understood. The fuel also would operate in an environment not greatly different from currently proposed gas cooled thermal reactors, and one which may well become commonplace in a few years' time.

It appears, therefore, that the gas cooled fast reactor does not present problems of a new kind. There are, however, differences in degree in two particular requirements which are well beyond the targets set in gas cooled thermal reactors. One of these is the level of irradiation in the fuel which is necessary to provide attractive fuel costs. At 10% burnup, this is several times as great as that in thermal reactors, and enters a range where new problems, such as fuel swelling, become significant. Fuel development for sodium cooled fast reactors may well provide much of the information required to design gas cooled fuel elements for the high burnup. No reason is seen at present for a major and expensive fuel development programme in addition to those already in hand for sodium cooled fast reactors and gas cooled thermal reactors though confirmatory tests under the particular conditions would be required before large scale construction were undertaken. Thus one can visualize the fuel concept for a gas cooled fast reactor emerging over the next few years. At the same time, it must be recognized that, at a later stage of development, there may be major differences in the form the fuel takes for gas and sodium cooled fast reactors. For example, the use of carbide fuel would introduce very different compatibility problems in the event of cladding failure.

A difference of degree also exists in the safety requirements of pressure vessels for fast and thermal gas cooled reactors. A preliminary assessment of depressurization incidents suggests that unless water spray injection for emergency cooling is resorted to, the maximum permissible hole area due to accident in a fast reactor vessel could be only about 5% or less of that acceptable in a currently typical thermal reactor. With close attention to the detail of all vessel penetrations, the required integrity may be achievable. It is basically a design problem rather than one that depends on a major programme of research, but much remains to be understood before the implications can be fully appreciated. With sufficient encouragement from experience of fast reactor fuel at long burnup, and from the continued experience with large concrete pressure vessels, the feasibility of the gas cooled fast reactor might become well enough established to justify an independent development programme directed towards proving its economic potential in comparison with sodium systems.

It seems that there would be a clear incentive to raise coolant pressures and fuel rating well above the figures suggested above, and a pressure of

140 atm might well be contemplated. Because the steam generating plant is likely to be very much larger than the reactor core, development to reduce its size, or to eliminate it, would also be indicated as well as continuing effort to ensure that undesirable effects do not result from tube failure. It is to be noted, however, that the immediately apparent advantage of the gas cooled fast reactor, its offer of a high breeding ratio, becomes economically less significant as a fast reactor construction programme proceeds, and depends on the existence of a plutonium shortage. There may be no case, therefore, for delaying the introduction of the system to await longer term developments of the type just mentioned above.

Even without the longer term developments in gas cooling, however, the development status relative to the sodium cooled fast reactor is such that the latter is likely to be available first by a margin of some years. The relative economics of the two systems as they will stand in say the 1980's is a matter for conjecture, and depends on how each develops over the years. The technology it shares with other reactor systems being developed gives the gas cooled reactor a particular interest, and indeed would reduce the overall cost of development should its feasibility and economic worth be established in the future.

5. The Steam Cooled Fast Reactor

Just as the gas cooled fast reactor can be regarded as an extension of the gas cooled thermal reactor, so the concept of a steam cooled fast reactor may be considered as a further development of the steam generating and boiling water thermal reactor technologies. Such a proposal has the potentialities of combining the capital cost savings of direct coupling to the steam turbine with the low fuel cost of the fast breeder reactor.

Within the framework of steam cooling, there are several basic considerations which influence the choice of the design concept. Of these, the choice of the operating coolant pressure is perhaps the most significant and a number of organizations throughout the world have studied both sub and supercritical pressure steam cooled fast reactors. Operation of the reactor at supercritical steam pressures, while offering the higher thermal efficiencies of the most advanced current steam plant, introduces challenging problems in the field of pressure vessel and coolant circuit design. Nevertheless, it is believed that with the developing techniques in pressure vessel manufacture, powers of at least 1000 MW(e) would be possible from a single reactor cooled with steam in the supercritical régime.

When hydrogen is present in a fast reactor, the softening of the neutron spectrum that results inevitably leads to a worsening of the neutron economy. The direct effects are to increase the capture-to-fission ratio of plutonium 239 and to reduce the fast fission bonus from plutonium 240 and uranium 238.

As a result, the breeding potential of this reactor type is inherently less than for a sodium cooled fast reactor of equivalent output.

The integration of a steam cooled reactor into a national power programme would depend on the overall economics of the reactor and, in particular, whether the potential savings in capital cost which can be achieved with a viable design combined with good fuel rating are sufficient to offset this reduction in breeding capability.

The ultimate breeding performance achievable in a practical steam cooled reactor is closely coupled to a number of features of the design including the heat removal mechanisms, fuel element practicability, reactivity control and safety features. The performance penalty on breeding which follows from the presence of the hydrogen moderator leads the designer to adopt fuel configurations with the lowest acceptable coolant fraction commensurate with acceptable heat removal performance. In the case of the supercritical pressure steam cooled reactor, a great deal of development requires to be carried out in order to establish techniques for obtaining low coolant volume fractions that can be guaranteed to be preserved in an environment of high temperature and large coolant pressure drop throughout long irradiations. Furthermore, it is possible that the increased cost of fabricating tightly packed fuel elements to close tolerances could outweigh the advantages, measured in terms of cycle efficiency, of supercritical over subcritical pressure steam.

With the highly rated fuel essential in a fast reactor, loss of coolant is an accident condition of crucial importance which requires attention from the outset of a conceptual design assessment. In the case of a steam cooled reactor, means must be found for either introducing emergency coolant within the individual fuel assemblies or of effectively nullifying the consequences of a failure of the pressure circuit.

The provision of control devices for the compensation of reactivity changes with burnup may influence appreciably the breeding ratio potential of the reactor. From this point of view, an important aspect of a conceptual study must be the proposed operation of the fuel cycle and the parasitic absorption needed to control core reactivity during the fuel cycle. The greater the control demanded of absorbers, the greater will be the loss to breeding potential.

In the area of fuel management, the steam cooled reactor poses special problems related to the core access. Unlike the sodium cooled reactor, the core is required to be contained within a high pressure envelope unless the alternative arrangement is adopted of perforating the fuel with a multiplicity of small diameter coolant tubes with complex header connexions outside the core limits (the shell-and-tube fuel element). Small batch refuelling schemes, needed to reduce the fuel cycle reactivity perturbations to a minimum, have associated with them real problems of fuel transfer through a high pressure envelope or of breaking into the header system of the tube through-fuel design. While such access may ultimately prove feasible, it would seem at

this time that such a feature would lead to some significant complication and with this an economic penalty. As a result of such considerations, the tendency is to favour fuel cycles involving complete or near-complete core replacement to minimize the occasions when the pressure system has to be opened. Good power distribution is important in order to permit a near uniform burnup of the fuel to be achieved, but, in addition, control devices must be incorporated and the effective breeding ratio would be improved if a method of control could be devised that did not involve parasitic absorption of neutrons.

Whilst the effective breeding ratio of a particular fast reactor design is an important parameter in the assessment of the status of the concept, the fissile inventory is equally important, for two reasons. First, the fissile doubling time of the reactor is closely related to inventory, and secondly, the rate of commissioning fast reactors given a particular stockpile of plutonium is also related to inventory. The fissile loading of the core depends on the detail physics of the configuration and also upon the fuel rating of the system. Three key features which limit the fuel rating of the steam cooled fast reactor can be identified as (1) the maximum fuel temperature that is acceptable coupled with the minimum size of fuel pin that is both technically and economically acceptable, (2) the maximum cladding temperature for satisfactory heat transfer to the coolant, and (3) the permissible coolant mass flow with the attendant problem of core pressure drop.

In the conceptual design studies carried out by the UKAEA, heat ratings in the region of 300 W/g of fuel ($\int Kd\theta = 50$) have been considered. In these arrangements, which have been based on oxide fuel in circular fuel pins, the maximum fuel centre temperature approaches the fuel melting point (approximately 2860°C). Considering the large volume increases of the fuel due to swelling resulting from high burnup (approximately 10% of heavy atoms), a large voidage requires to be provided. This might be achieved by using sintered annular fuel pellets, or alternatively, low density, vibro-compacted fuel. At the present time there is a notable lack of information on the behaviour of the fuel under the relevant conditions of high burnup and temperature. Furthermore, experimental data are required on the compatibility of oxide fuels with high pressure steam, which is of importance in the case of cladding failure. Nitride and mixed nitride/carbide may be alternatives to oxide in view of their much higher conductivity which would afford substantial reduction in fuel temperatures, but their compatibility with steam requires experimental investigation.

The choice of cladding material is governed by the accepted criteria that it must possess adequate corrosion resistance and acceptable mechanical properties throughout irradiation. The cladding needs to have high yield strength and good resistance to creep. Typical operating conditions for a reactor cooled by supercritical steam (outlet temperature 565°C) are:

Maximum hot-spot can temperature	650–700°C
Coolant pressure	275 kg/cm ²
Dwell time	270 days
Maximum surface heat flux	270 kcal/h.cm ²

To maintain the principle of strong cladding, i.e. strong in yield and creep, for the operating conditions envisaged, it appears that nickel based alloys could be the material selected for use up to 700°C, whilst austenitic stainless

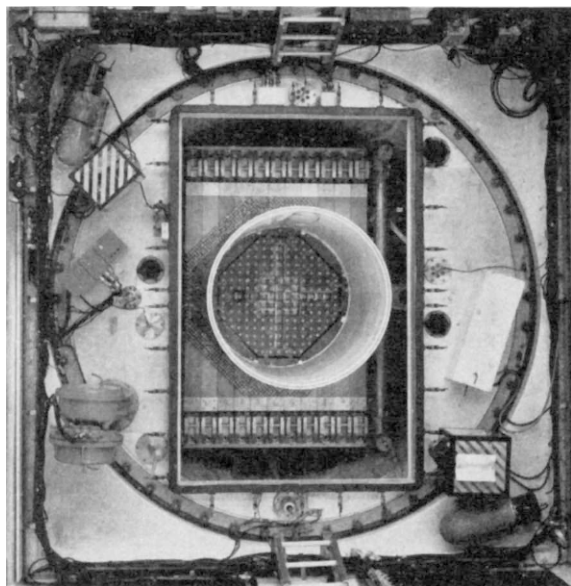


FIG. 3. A typical steam cooled fast reactor core simulation in Dimple at AEE, Winfrith.

steels could be considered for use up to 650°C. Existing data on the loss of strength and ductility under irradiation are scarce. Some of the nickel alloys and austenitic steels show good corrosion resistance based on static corrosion experiments, but further data are necessary to determine dynamic corrosion effects. Study is also required on the application of weak cladding to circular fuel pin designs, and on alternative fuel configurations, for example, plates.

The safety and dynamic characteristics of the reactor are features of overriding importance in the search for a completely balanced design. The reactivity effects, resulting from variations in the coolant density, need detailed study since these are important from the points of view of overall plant stability and ultimate accident considerations. The complete assessment of these effects is made complex by close coupling of different parts of the reactor system; notably the fuel region, the blankets and reactivity control features. UKAEA experience has indicated that sometimes misleading trends are

obtained as a result of omitting important physical effects from the model used in the preliminary assessment, for example it is important to allow in the reactor physics calculations for the change in reflector saving with change of coolant density. A concentrated effort in this field is required to identify a concept that is both safe and dynamically stable and which, at the same time, retains a satisfactory economic performance.

In parallel with these studies further examination of the nuclear data used in the physics calculation must continue and reactor physics experiments must be undertaken which are designed to identify the shortcomings of current nuclear data and available calculation techniques. Figure 3 shows such a physics experiment in the Dimple reactor at AEE Winfrith, using a mixed critical core. This consists of a central region of the same composition as a typical supercritical pressure steam cooled fast reactor surrounded by a thermal reactor driver region moderated by light water. The use of this type of arrangement should make it possible to measure sufficient detailed reaction rates to reduce the present major uncertainties in physics data whilst using an order of magnitude less fuel than required for a full-scale critical assembly.

In several countries assessment studies of steam cooled fast reactors are being carried out, and it is important to gain from these an appreciation of the true position of the steam cooled fast reactor. Before final assessments of the comparative status of this system can be made, however, it is apparent that much work in the fields of fuel technology and reactor physics remains to be done.

6. Fluid Fuel Fast Reactors

About 10 years ago, fluid fuel fast reactors attracted attention. It was considered that these could possess the advantages of being able to dump fuel to safe storage in the event of a fault and of sustaining higher fuel ratings than reactors with solid fuels. It was also postulated that limitations associated with burnup in the normal sense would not arise, and that avoiding fuel fabrication should have a substantial effect on fuel costs. Studies of such concepts based on fuel dispersed in liquid metals were undertaken at AERE, Harwell, between 1956 and 1959. These studies included plutonium-in-bismuth and the plutonium-magnesium eutectic. A third alternative was offered by the alloys of plutonium with iron, nickel, and cobalt and work in this direction at Los Alamos (U.S.A.) went as far as a 1 MW experiment (LAMPRE). However, none of the many ideas considered resulted in a sufficiently good combination of economic incentive and likelihood of successful development for them to be further pursued. It is not now believed that a significant amount of work is anywhere in progress on such systems.

More recently, molten salt as a fluid fuel has come into prominence as a result of the Oak Ridge National Laboratory (ORNL) work on thermal fused

salt systems. Fast reactors using molten salt fuel have been studied by the UKAEA, an assessment having been made of a fast system operating on a plutonium/uranium 238 cycle using sodium chloride as the carrier salt. To avoid waste of fissile material such as would result from the large out-of-core inventory of an externally cooled system or the poor neutron economy associated with a design having coolant tubes situated within the core, the direct-contact coolant system originally suggested by ORNL has been

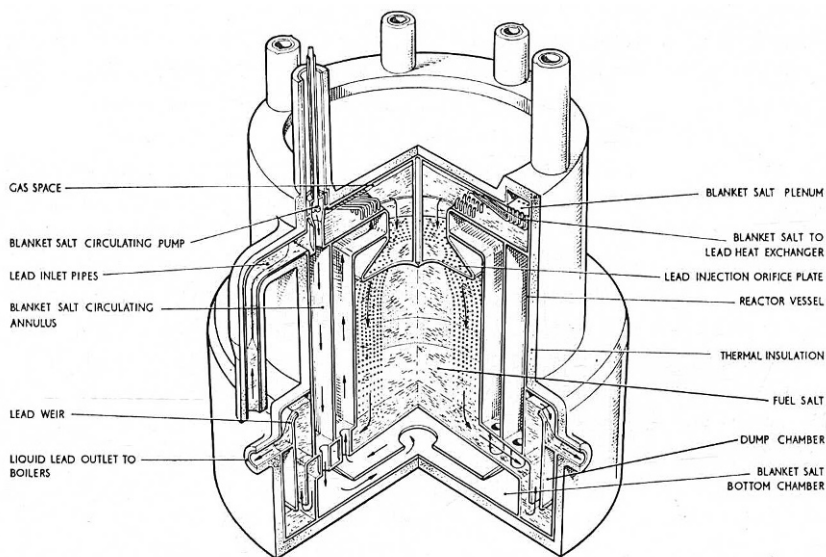


FIG. 4. Possible arrangement of a molten salt reactor.

examined, the molten salt being cooled and circulated within the reactor vessel by entrainment in a stream of molten lead droplets. These are injected at the top of the vessel and form a curtain in the outer part of the core region (Fig. 4). The lead separates from the fuel salt and is circulated through a steam generator. This arrangement results in a very compact design though care must be taken to leave sufficient space for remote maintenance. The system has a number of attractive safety features; it has a large prompt negative temperature coefficient, and coolant failure renders the core sub-critical, while there is no equivalent to the meltdown accident.

The system has potential attractions but in view of the amount of new technology involved, the first stage in the development of a viable molten salt fast reactor would have to be a programme of experimental work extending over several years to establish the basic feasibility of the scheme. The study conducted by the UKAEA has indicated those areas in which technical information is required and has suggested lines along which solutions

could reasonably be sought. In addition to certain chemical processing problems, the primary uncertainties lie in the choice of materials for the reactor vessel which would both resist corrosion and be capable of fabrication to the desired high standards and in the heat transfer from the core salt to the lead curtain.

7. Conclusions

The paper has reviewed the status of fast breeder reactor technology essentially on the basis of work undertaken by the UKAEA. The U.K.'s requirements for electric power are such that there is a premium on introducing the fast reactor at the earliest possible date. This situation has been recognized for some time and the steps that were taken during the 1950's have provided this country with the ability to undertake the construction of a large-scale prototype fast reactor power plant in which sodium is being used as the reactor coolant. The attainment of such a stage in fast reactor technology implies a notable accumulation of design and operating experience in liquid metal coolant techniques and the construction and operation of the Dounreay fast reactor experiment has made a considerable contribution in these areas.

However, at this stage in fast reactor development, it would be unwise to assume that any one type of system constitutes the ultimate of attainment and the UKAEA along with many other similar organizations throughout the world has devoted effort to exploring the potential of other types of fast reactor. It is too early to judge how these stand relative to the sodium cooled system or even how far along its development path the sodium cooled reactor has advanced. Much work in the fields of reactor physics, fuel and cladding technology and materials evaluation is needed before answers to these questions can be formed with certainty, but it is clear that the potential of the fast reactor principle is so great that this effort must be expended to enable the world power requirements of the future to be met.

8. Acknowledgements

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